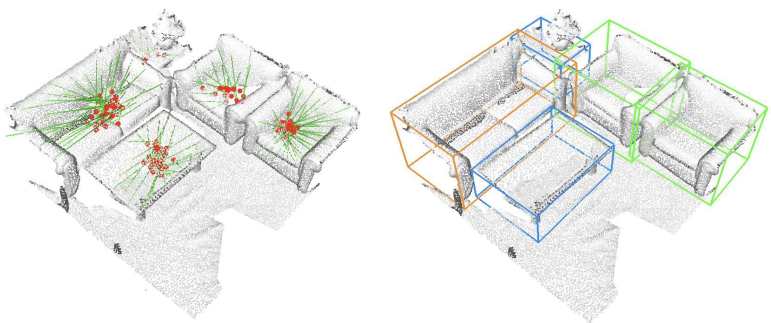
Deep Learning for Point Cloud Data

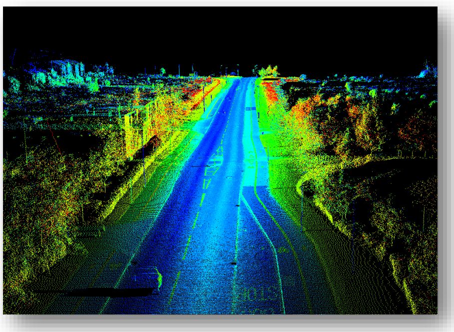
thrive SIGGRAPH2019
LOS ANGELES • 28 JULY – 1 AUGUST



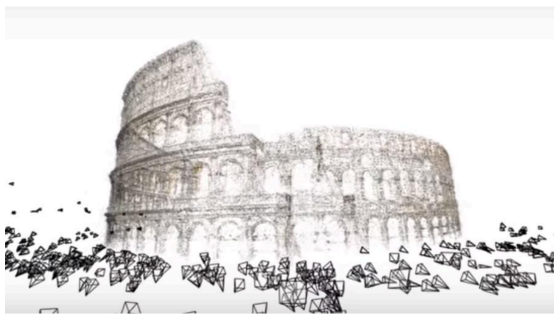
Leonidas Guibas
Stanford University
Facebook AI Research

Leonidas Guibas Laboratory
Geometric Computing

Point Clouds



Lidar point clouds (LizardTech)

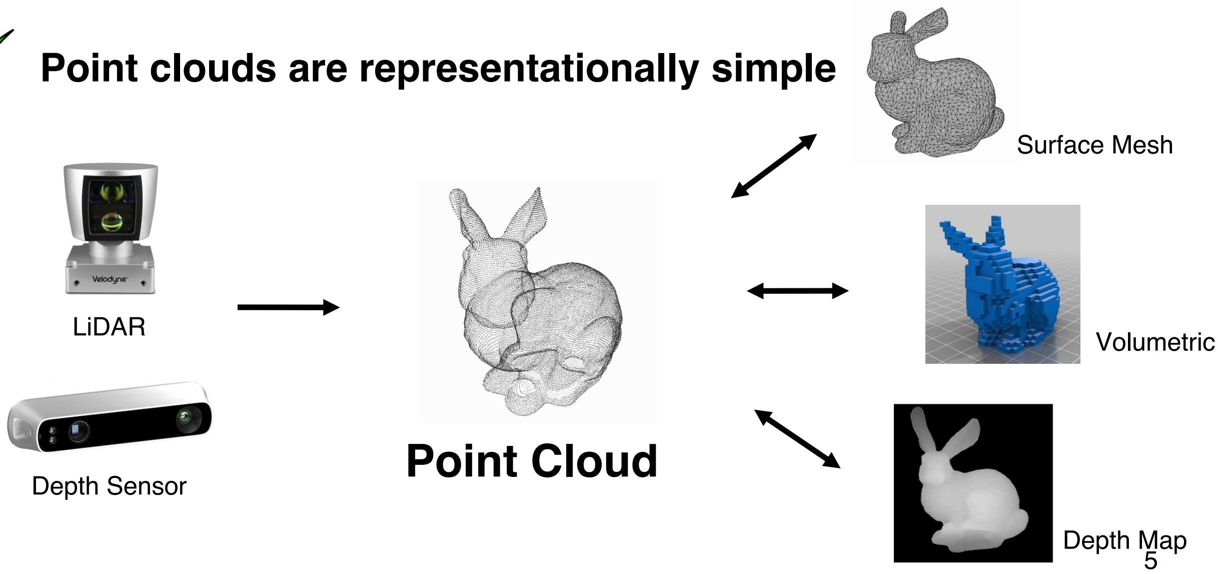


Structure from motion (Microsoft)

Depth camera (Intel)

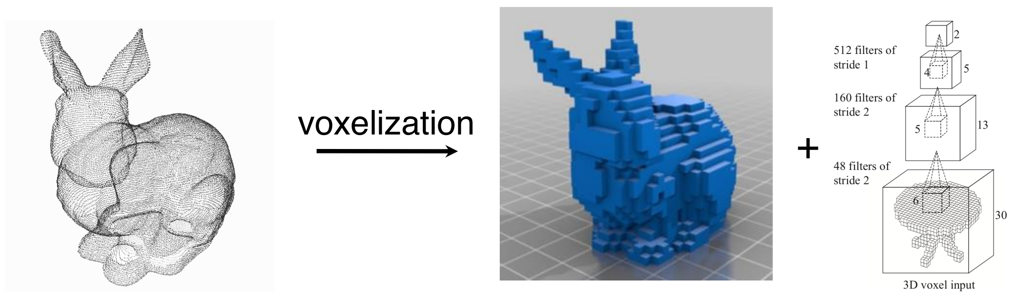
A Common 3D Representation: Point Cloud

- **Point clouds are close to raw sensor data**
- **Point clouds are representationally simple**



Early Work on 3D Learning

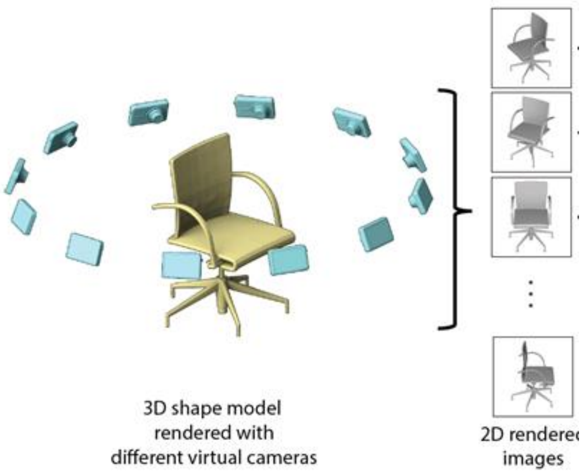
Point clouds were **converted to other regular representations** before input to a deep neural network



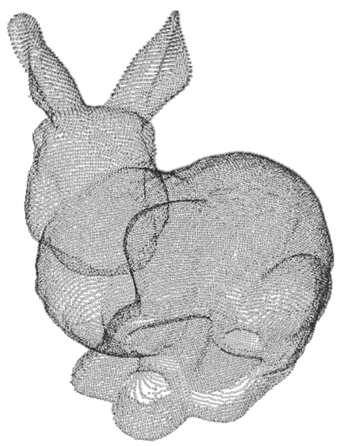
Con: High space & time complexity -- 3D convolution $O(N^3)$
Quantization errors in voxelization

Earlier Work

Point clouds were **converted to other regular representations** before input to a deep neural network



Multiview Images Implicit Models



Research Question:

Can we achieve effective **feature learning directly on irregular point clouds**?

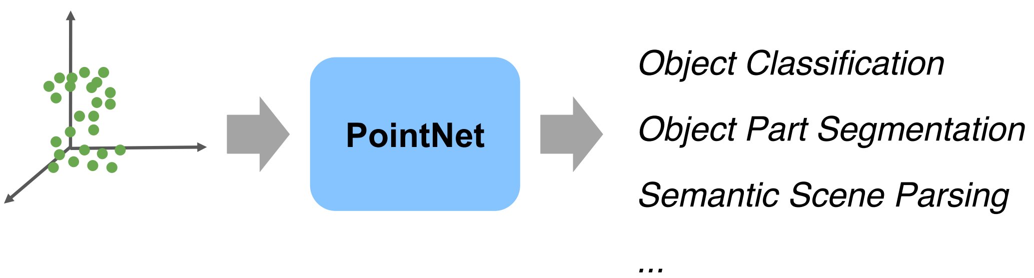
Talk Ouline

- Survey of PointNet, PointNet++ architectures (~2017)
- Since the original PointNet work, an explosion of activity in this area -- very brief survey
- Point Cloud Synthesis

PointNet Architecture Review

End-to-end learning for irregular point data

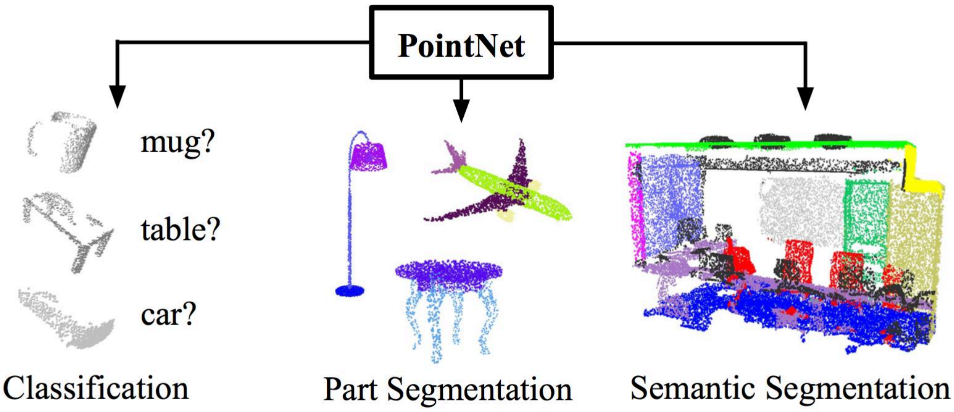
Unified framework for various tasks



PointNet Architecture Review

End-to-end learning for irregular point data

Unified framework for various tasks



Challenges

The model has to respect key properties of point clouds:

Point Permutation Invariance

Point cloud is a set of unordered points

Spatial Transformation Invariance

Point cloud rigid motions should not alter classification results

Challenges

The model has to respect key properties of point clouds:

Point Permutation Invariance

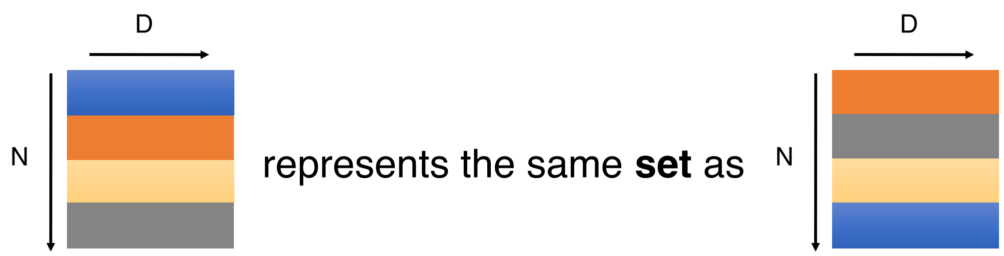
Point cloud is a set of unordered points

Spatial Transformation Invariance

Point cloud rigid motions should not alter classification results

Unordered Input

Point cloud: set of N **unordered** points, each represented by a D dim vector



Model needs to be invariant to N! permutations

Permutation Invariance: Symmetric Function

$$f(x_1, x_2, \ldots, x_n) \equiv f(x_{\pi_1}, x_{\pi_2}, \ldots, x_{\pi_n}), \quad x_i \in \mathbb{R}^D$$

Examples:

$$f(x_1, x_2, \ldots, x_n) = \max\{x_1, x_2, \ldots, x_n\}$$
$$f(x_1, x_2, \ldots, x_n) = x_1 + x_2 + \ldots + x_n$$

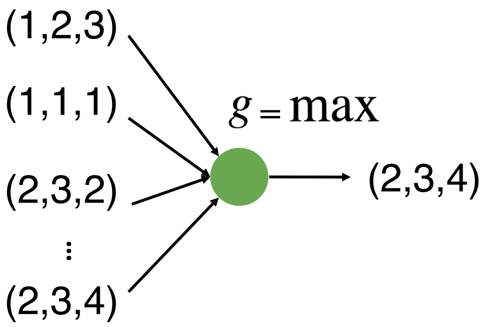
…

How can we construct a universal family of symmetric functions by neural networks?

Construct Symmetric Functions by Neural Networks

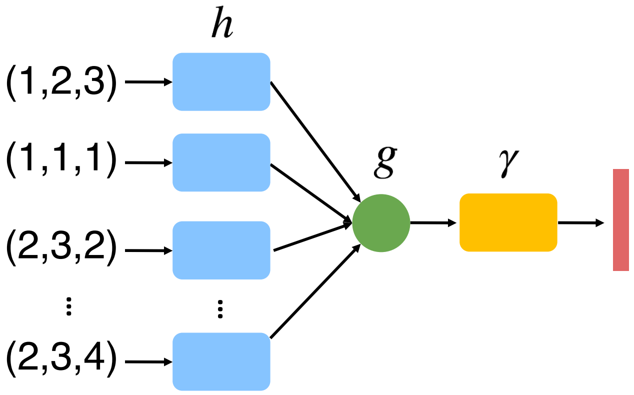
Simplest form: directly aggregate all points with a symmetric operator g

Just discovers simple extreme/aggregate properties of the geometry.



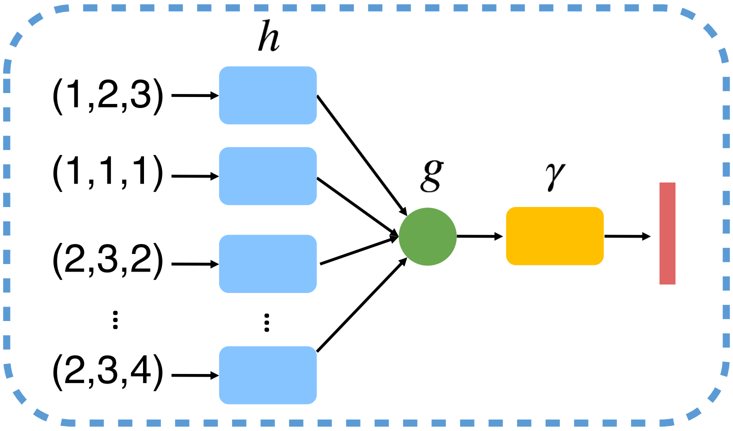
Construct Symmetric Functions by Neural Networks

Embed points to a high-dim space before aggregation.
Aggregation in the (redundant) high-dim space encodes more interesting properties of the geometry.



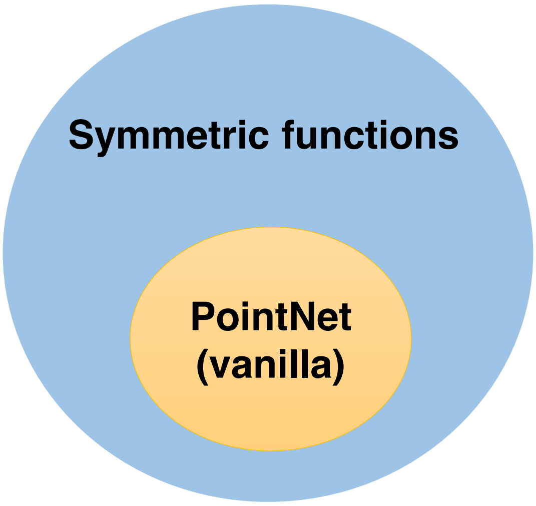
Construct Symmetric Functions by Neural Networks

$f(x_1, x_2, \ldots, x_n) = \gamma \circ g(h(x_1), \ldots, h(x_n))$ is symmetric if g is symmetric



PointNet (vanilla)

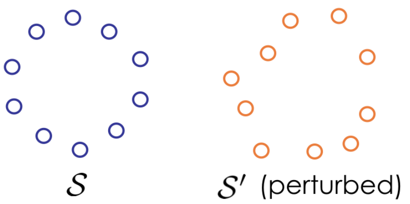
What Symmetric Functions Can Be Constructed By PointNet?



PointNet as a Universal Approximation to Set Functions

Hausdorff continuous:

$f: 2^{\mathcal{X}} \rightarrow \mathbb{R}$ is a continuous set function w.r.t Hausdorff distance



if $d_{Hausdorff}(S, S') \approx 0$, then $f(S) \approx f(S')$

Theorem

A Hausdorff continuous set function $f: 2^{\mathcal{X}} \rightarrow \mathbb{R}$ can be arbitrarily approximated by PointNet.

$$\left| f(S) - \boxed{\gamma \left(\underset{x_i \in S}{\mathrm{MAX}} \{h(x_i)\} \right)} \right| < \epsilon$$

$S \subseteq R^d$

PointNet (vanilla)

Voxel occupancy maps

Challenges

The model has to respect key properties of point clouds:

Point Permutation Invariance

Point cloud is a set of unordered points

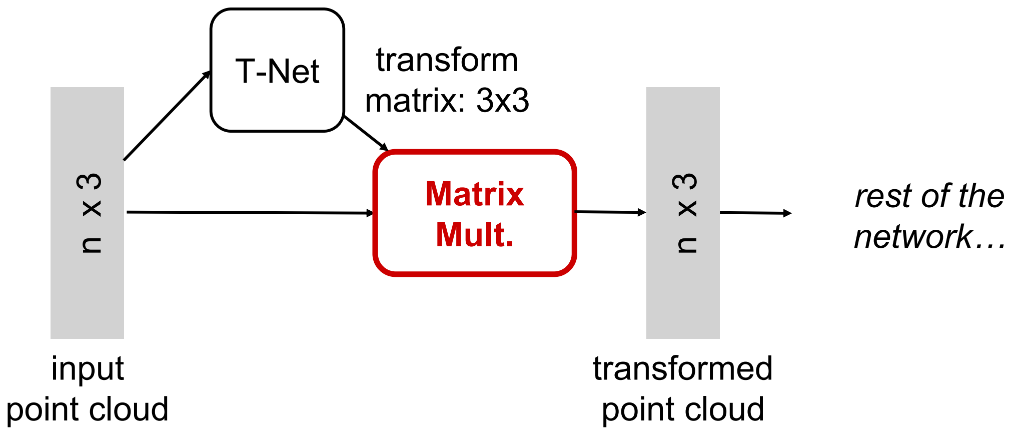
Transformation Invariance

Point cloud rigid motions should not alter classification results

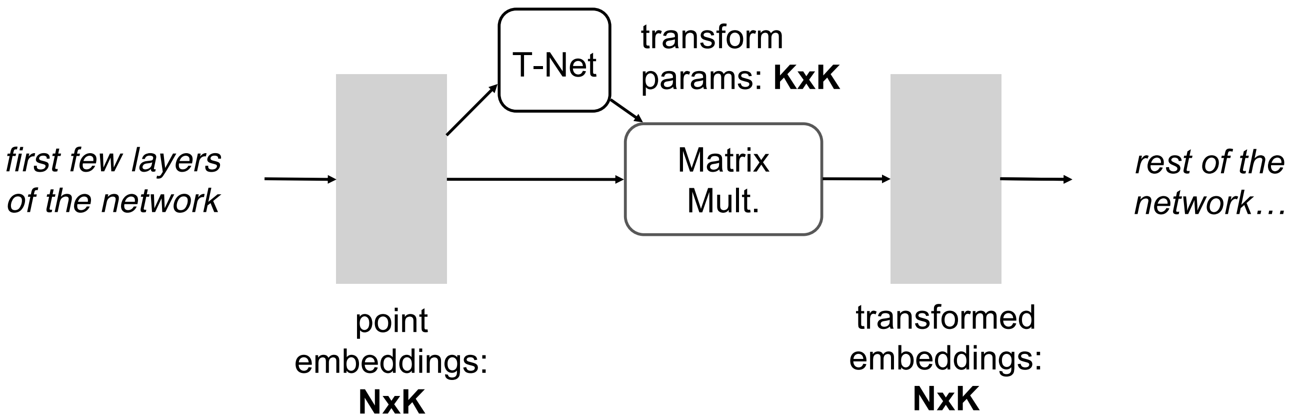
Input Alignment by Transformer Network

Idea: Data dependent transformation for automatic alignment

The transformation is just matrix multiplication!



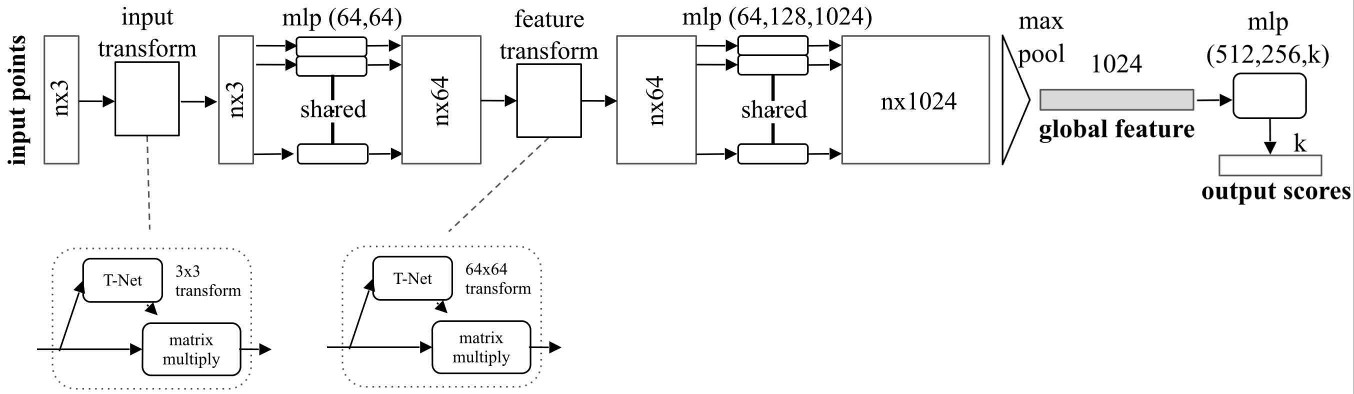
Embedding Space Alignment



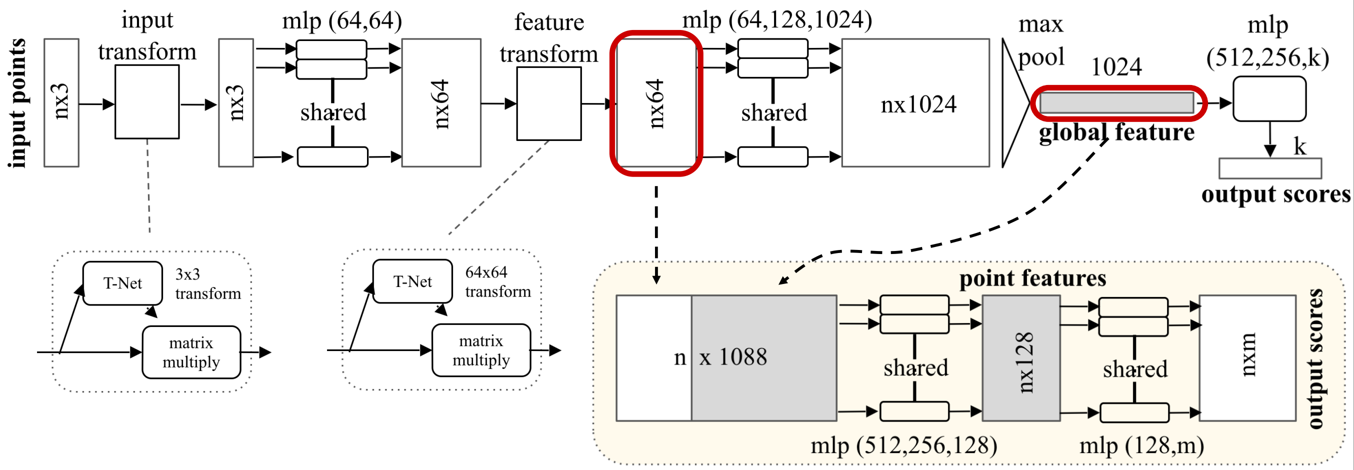
Regularization loss:
Transform matrix close to orthogonal: $L_{reg} = \|I - AA^T\|_F^2$

Point Classification and Segmentation

PointNet Classification Network



Extension to PointNet Segmentation Network



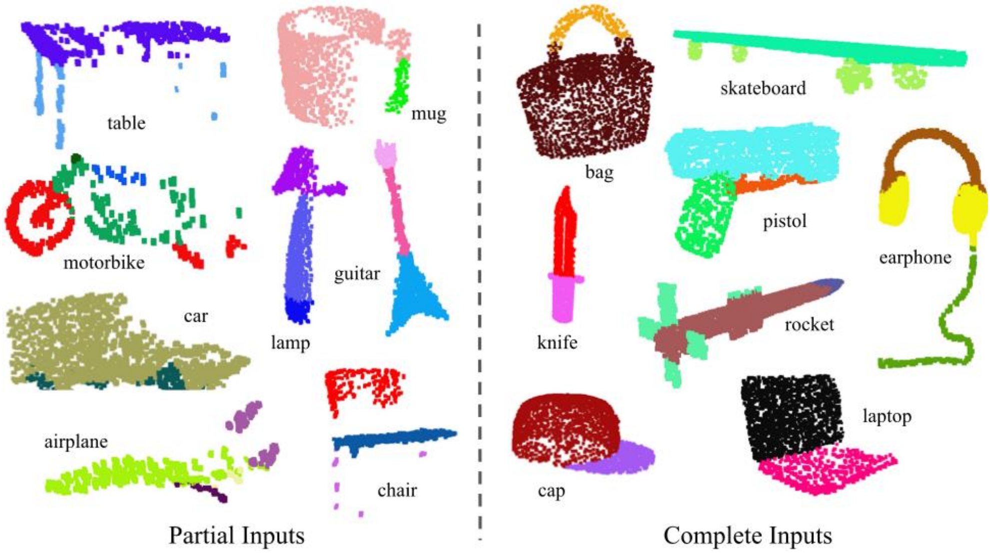
Results on Object Classification

	input	#views	accuracy avg. class	accuracy overall
SPH [12]	mesh	-	68.2	
3DShapeNets [29]	volume	1	77.3	84.7
VoxNet [18]	volume	12	83.0	85.9
Subvolume [19]	volume	20	86.0	**89.2**
LFD [29]	image	10	75.5	-
MVCNN [24]	image	80	**90.1**	-
Ours baseline	point	-	72.6	77.4
Ours PointNet	point	1	86.2	**89.2**

3D CNNs

dataset: ModelNet40; metric: 40-class classification accuracy (%)

Results on Object Part Segmentation

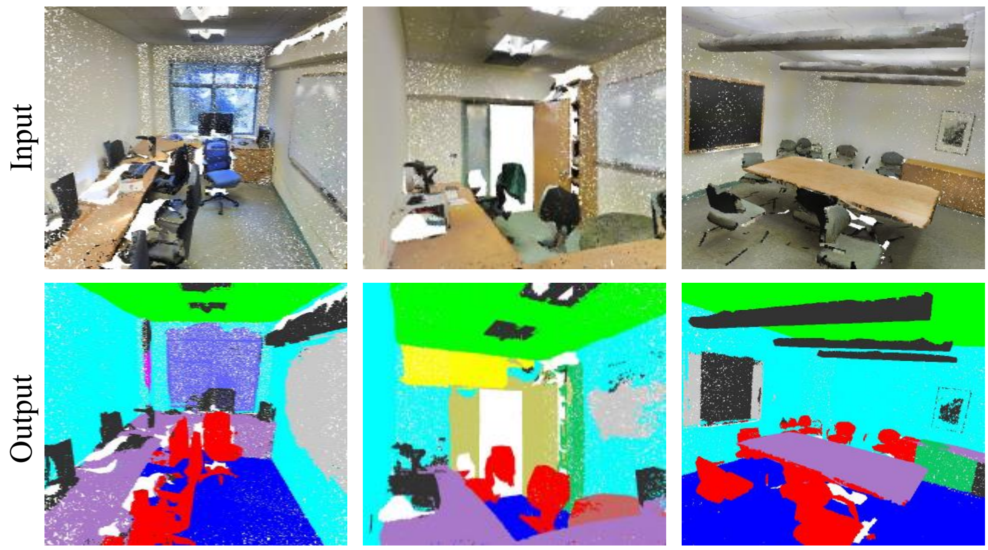


Results on Object Part Segmentation

	mean	aero	bag	cap	car	chair	ear phone	guitar	knife	lamp	laptop	motor	mug	pistol	rocket	skate board	table
# shapes		2690	76	55	898	3758	69	787	392	1547	451	202	184	283	66	152	5271
Wu [28]	-	63.2	-	-	-	73.5	-	-	-	74.4	-	-	-	-	-	-	74.8
Yi [30]	81.4	81.0	78.4	77.7	**75.7**	87.6	61.9	**92.0**	85.4	**82.5**	**95.7**	**70.6**	91.9	**85.9**	53.1	69.8	75.3
3DCNN	79.4	75.1	72.8	73.3	70.0	87.2	63.5	88.4	79.6	74.4	93.9	58.7	91.8	76.4	51.2	65.3	77.1
Ours	**83.7**	**83.4**	**78.7**	**82.5**	74.9	**89.6**	**73.0**	91.5	**85.9**	80.8	95.3	65.2	**93.0**	81.2	**57.9**	**72.8**	**80.6**

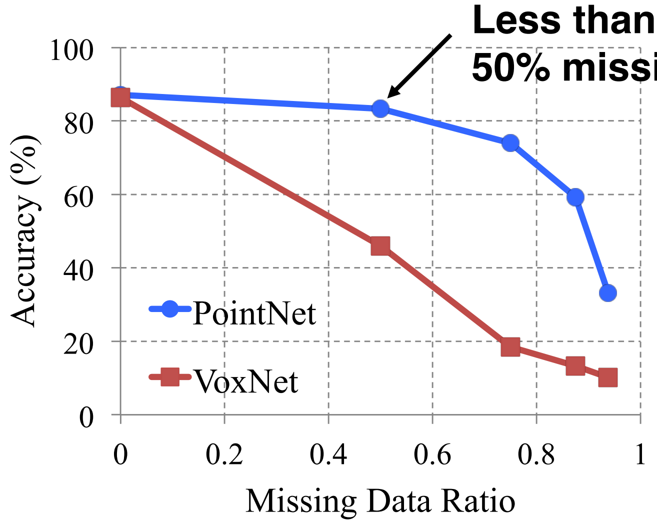
dataset: ShapeNetPart; metric: mean IoU (%)

Results on Semantic Scene Parsing



dataset: Stanford 2D-3D-S (Matterport scans)

PointNet is Robust to Data Corruption



Less than 2% accuracy drop with 50% missing data

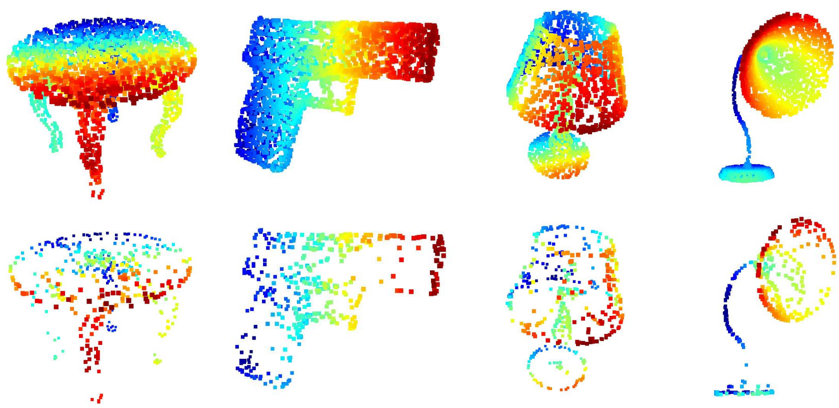
Why is PointNet so robust to missing data?

dataset: ModelNet40; metric: 40-class classification accuracy (%)

Visualizing Global Point Cloud Features

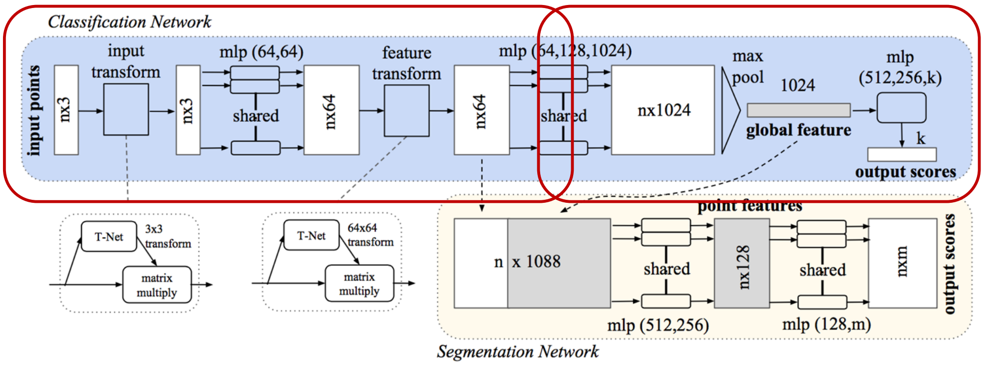
Original Shape

Critical Points



PointNet learns to pick perceptually interesting points!

Learning Interesting Points

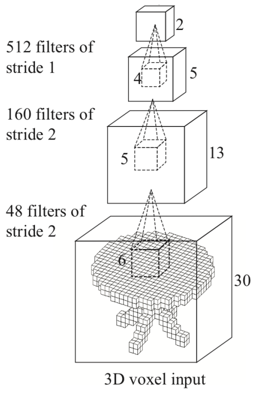


- Pointnet learns optimization criteria, which in turn pick interesting points

From PointNet to PointNet++

Limitations of PointNet

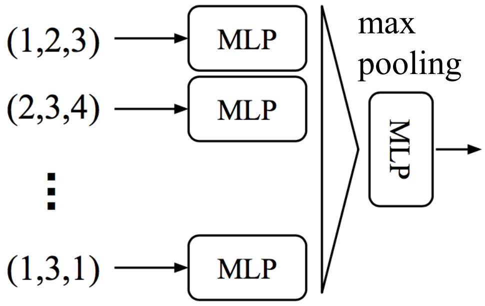
Hierarchical feature learning
multiple levels of abstraction



3D CNN [Wu et al.2015]

v.s.

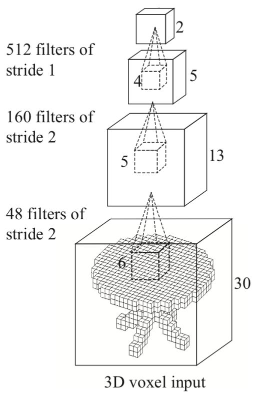
Global feature learning
either **one** point or **all** points



PointNet (vanilla) [Qi et al.2017]

Limitations of PointNet

Hierarchical feature learning
multiple levels of abstraction



3D CNN [Wu et al.2015]

v.s.

Global feature learning
either **one** point or **all** points

No local context

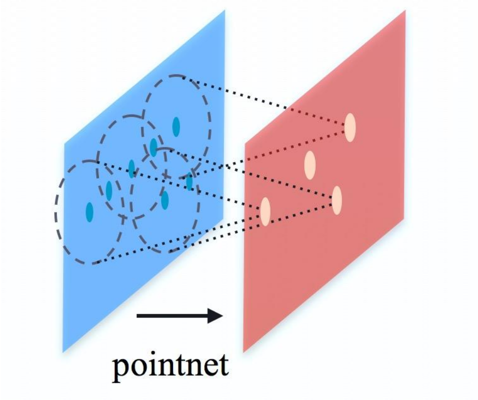
Limited local invariance

PointNet (vanilla) [Qi et al.2017]

PointNet++

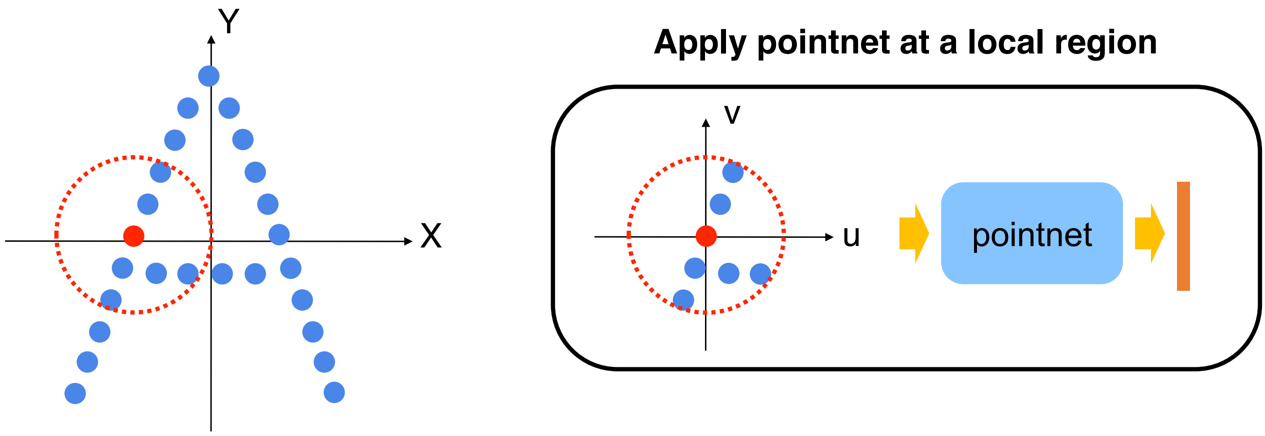
Basic idea: Recursively apply pointnet at local regions.

✓ Hierarchical feature learning
✓ Local translation invariance
✓ Permutation invariance



[2] *Charles R. Qi, Li Yi, Hao Su, Leonidas Guibas. PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space (NIPS'17)*

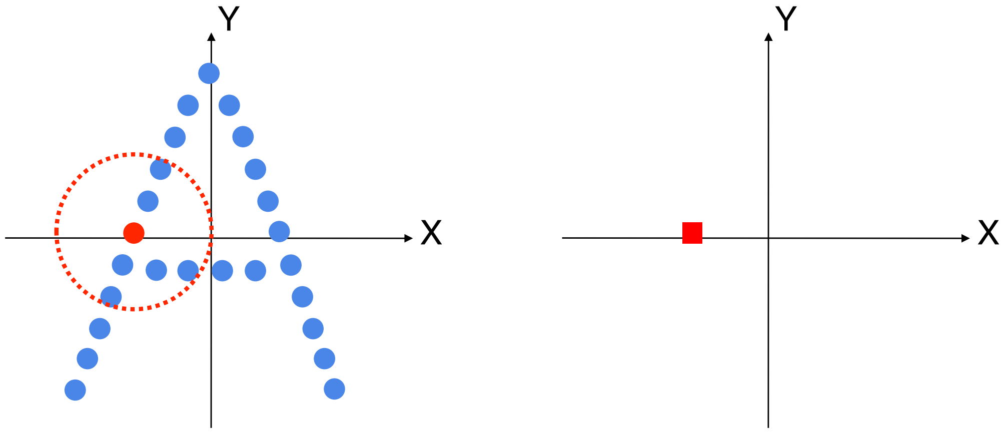
Hierarchical Point Feature Learning



N points in (X,Y)

k points in local coordinates (u,v)

Hierarchical Point Feature Learning



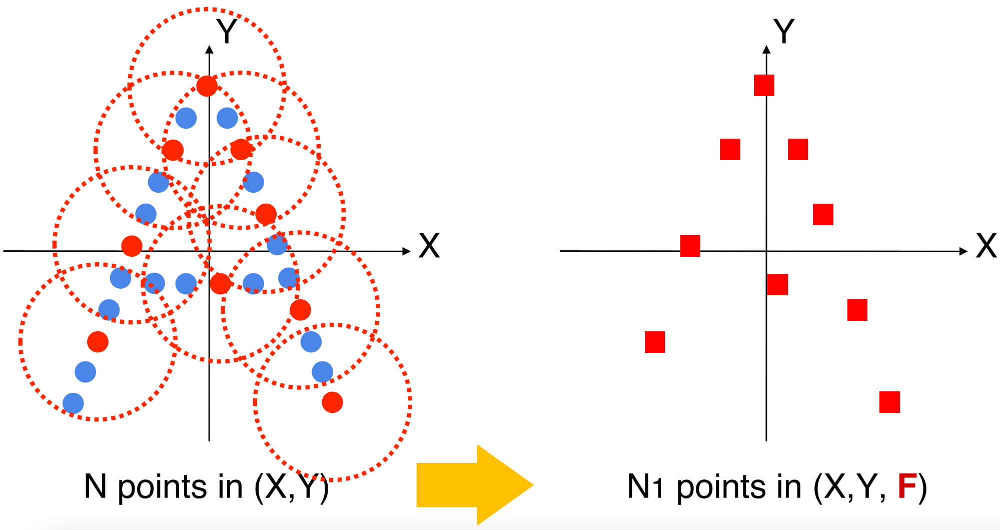
N points in (X,Y)

points in (X,Y, **F**)

Euclidean space

high-dim feature space

Hierarchical Point Feature Learning

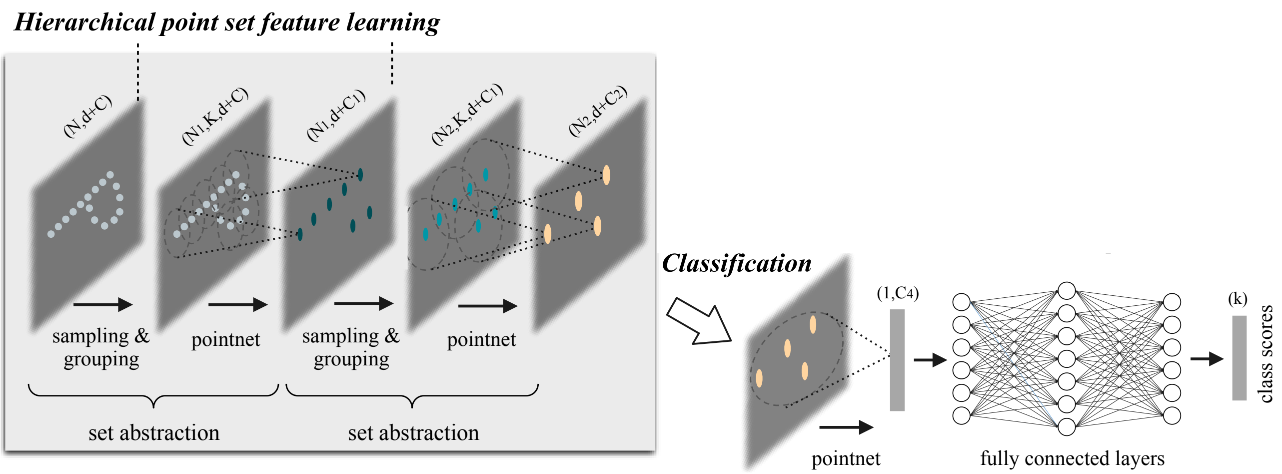


N points in (X,Y)

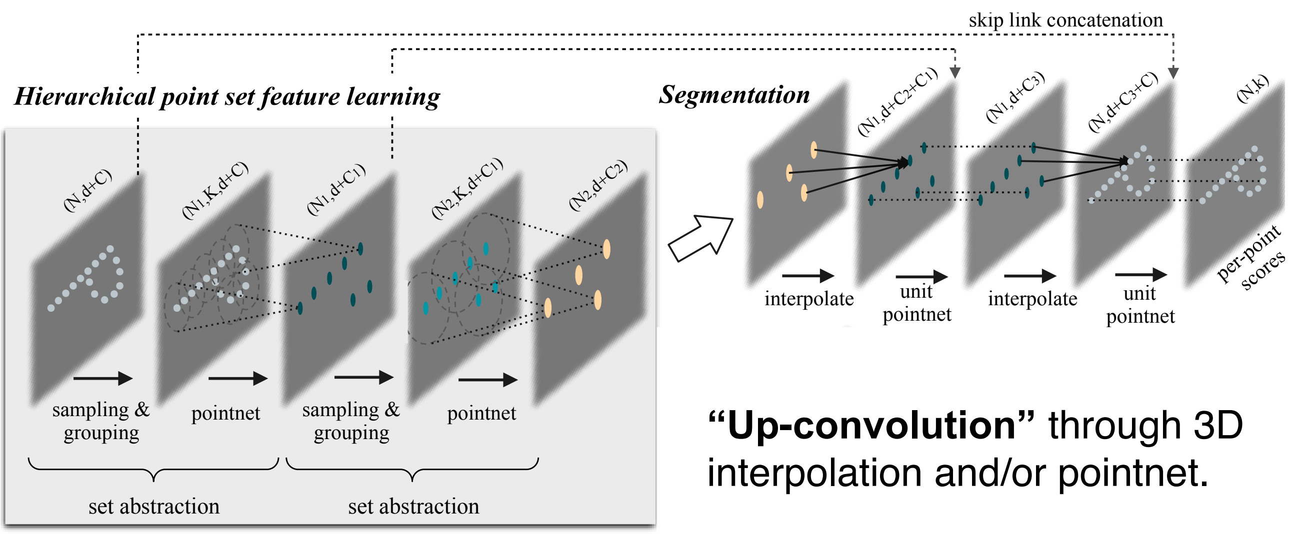
N_1 points in (X,Y, **F**)

Set Abstraction: farthest point sampling + grouping + pointnet

PointNet++ for Classification and Segmentation



PointNet++ for Classification and Segmentation

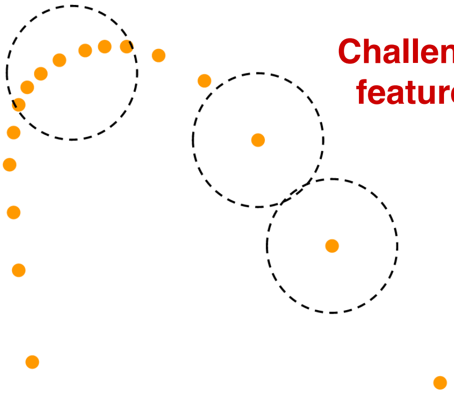


“Up-convolution” through 3D interpolation and/or pointnet.

Non-uniform Sampling Density in Point Clouds

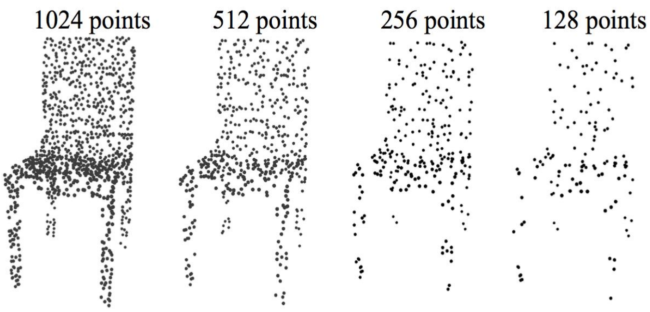
Density variation is a common issue in 3D point cloud processing

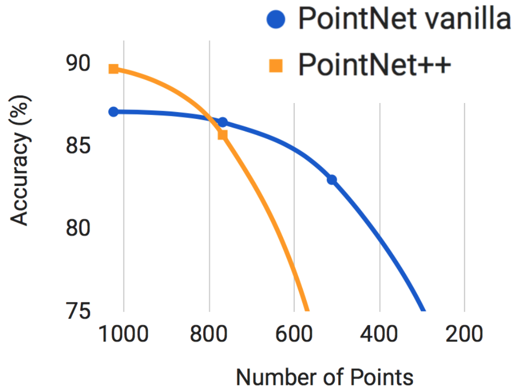
- perspective effect, radial density variation, motion etc.



Challenge for local feature learning!

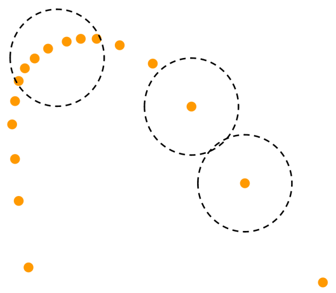
Density Variation Affects Hierarchy

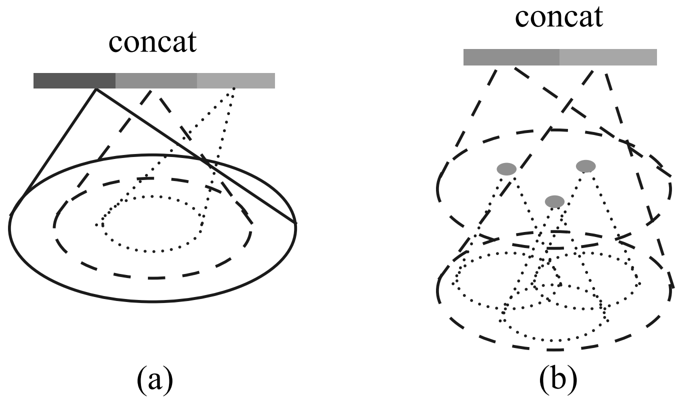




Small kernels suffer from varying densities!

Robust Learning Under Varying Sampling Density

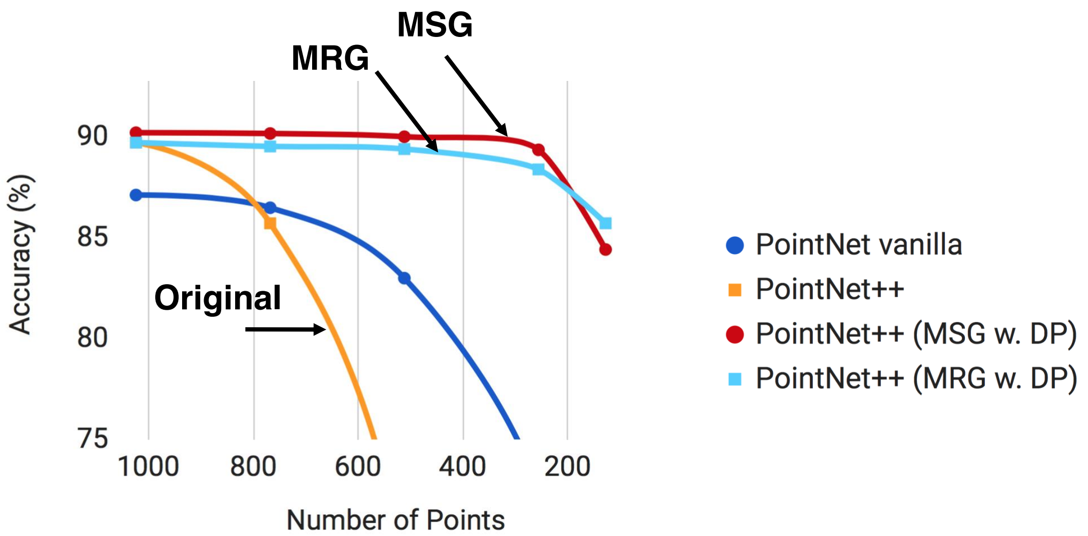




Multi-scale grouping (MSG) Multi-res grouping (MRG)

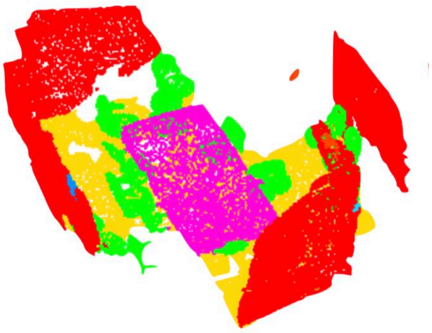
During Training: input point dropout with random dropout ratio

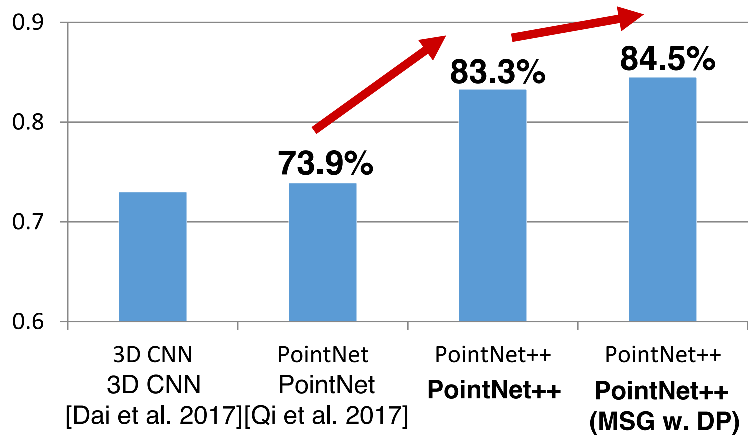
Robust Learning Under Varying Sampling Density



PointNet++ Results: Scene Parsing

Better accuracy with hierarchical learning.

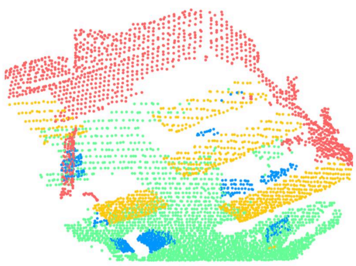


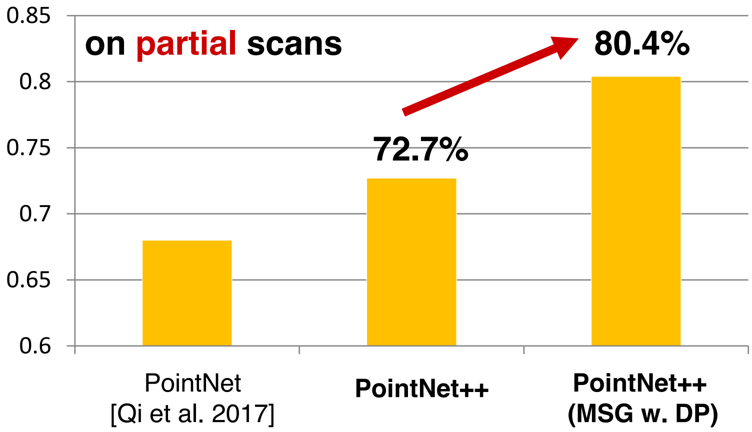


dataset: ScanNet; metric: per-point semantic classification accuracy (%)

PointNet++ Results: Scene Parsing

Robust layers for non-uniform densities (MSG) helps a lot.





dataset: ScanNet; metric: per-point semantic classification accuracy (%)

PointNet++ Results: Non-Euclidean Space

For organic shape recognition, PointNet++ can generalize to non-Euclidean space

- ❖ intrinsic point features (HKS, WKS, Gaussian curvature)
- ❖ intrinsic distance metric (geodesic)

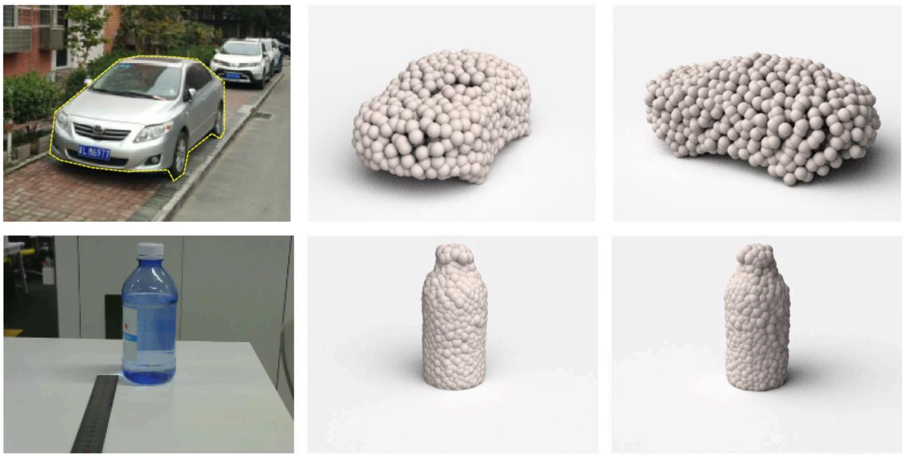
(a) Horse (b) Cat (c) Horse

	Metric space	Input feature	Accuracy (%)
DeepGM [13]	-	Intrinsic features	93.03
Ours	Euclidean	XYZ	60.18
	Euclidean	Intrinsic features	94.49
	Non-Euclidean	Intrinsic features	**96.09**

dataset: SHREC15; metric: shape classification accuracy (%)

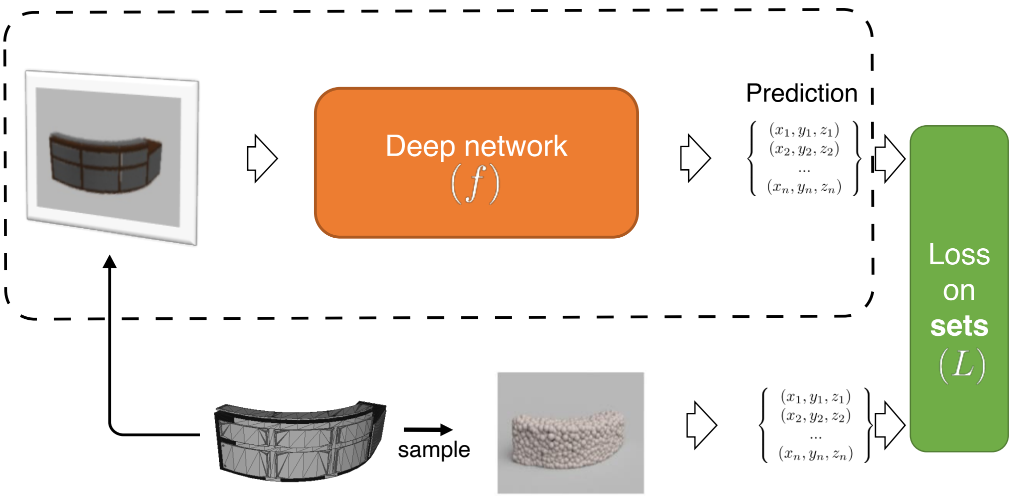
Point Cloud Synthesis

Point Cloud Synthesis from a Single Image

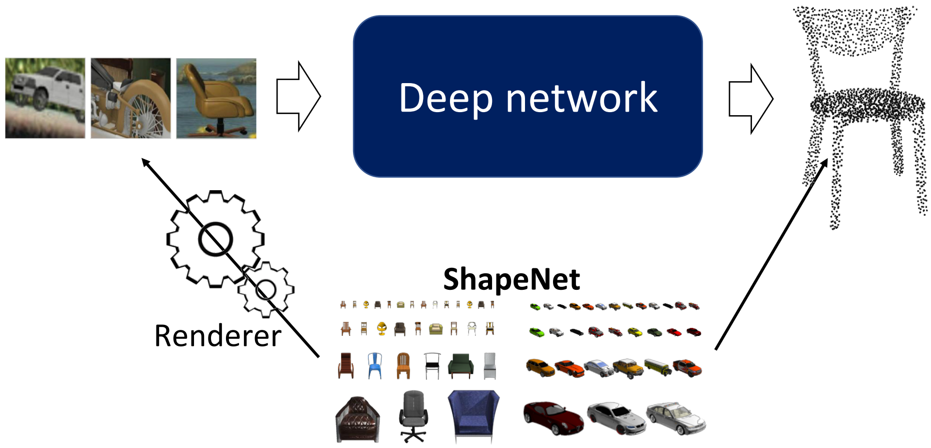


Hao Su, Haoqiang Fan, Leonidas Guibas
Learning Shape Abstractions by Assembling Volumetric Primitives
CVPR 2017

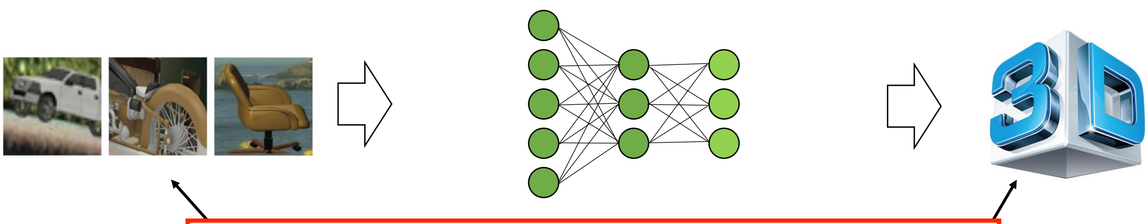
End-to-End Learning



Synthesize for Learning



Supervision from “Synthesize for Learning”



- **200K shapes from 2K categories**
- **10M images with ground truth**

Point Cloud Distance Metrics

$\boxed{\text{Worst case:}}$ Hausdorff distance (HD)

$$d_{\mathrm{H}}(X,Y) = \max\left\{ \sup_{x\in X}\inf_{y\in Y} d(x,y),\ \sup_{y\in Y}\inf_{x\in X} d(x,y) \right\},$$

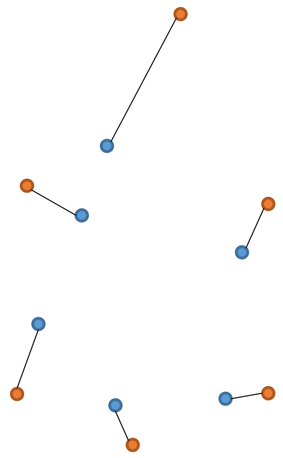
$\boxed{\text{Average case:}}$ Chamfer distance (CD)

$$\mathrm{CD}(S_1,S_2) = \frac{1}{|S_1|}\sum_{x\in S_1}\min_{y\in S_2}\|x-y\|_2^2 + \frac{1}{|S_2|}\sum_{y\in S_2}\min_{x\in S_1}\|x-y\|_2^2$$

$\boxed{\text{Optimal case:}}$ Earth Mover's distance (EMD)

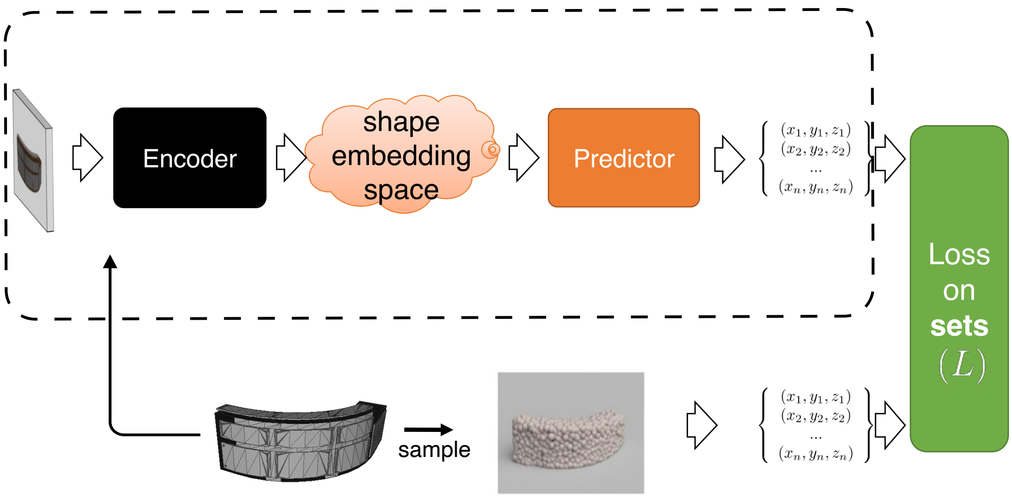
$$d_{EMD}(S_1,S_2) = \min_{\phi:S_1\to S_2}\sum_{x\in S_1}\|x-\phi(x)\|_2$$

where $\phi : S_1 \to S_2$ is a bijection.

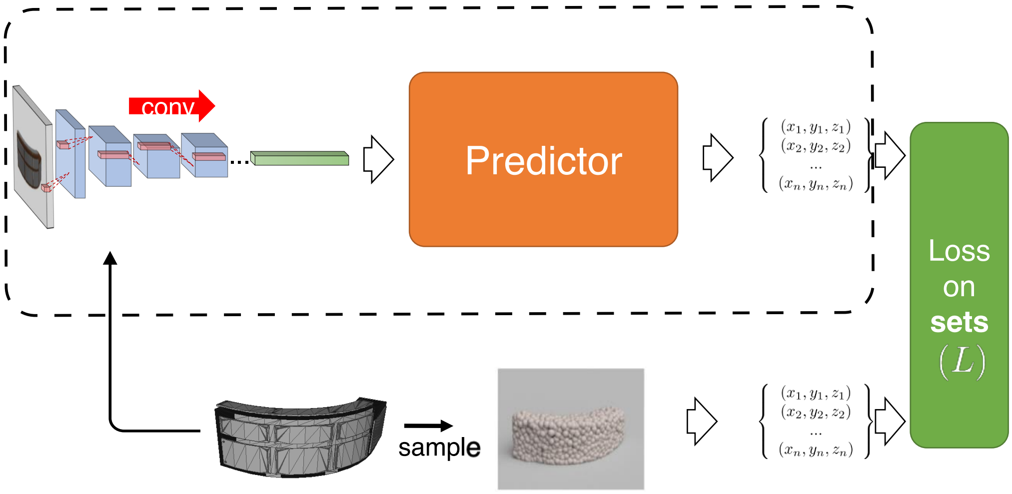


Solves the optimal transportation (bipartite matching) problem!

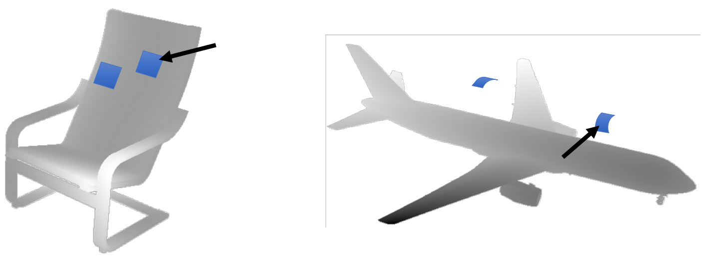
End-to-End Learning



End-to-End Learning

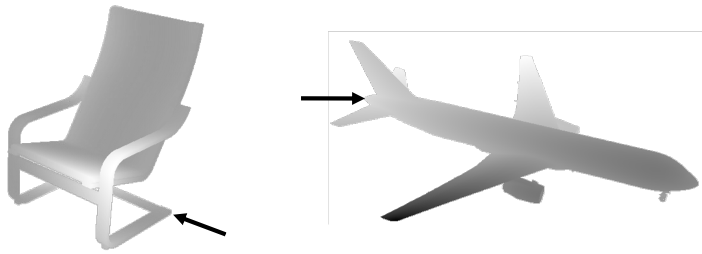


Natural Statistics of Object Geometry



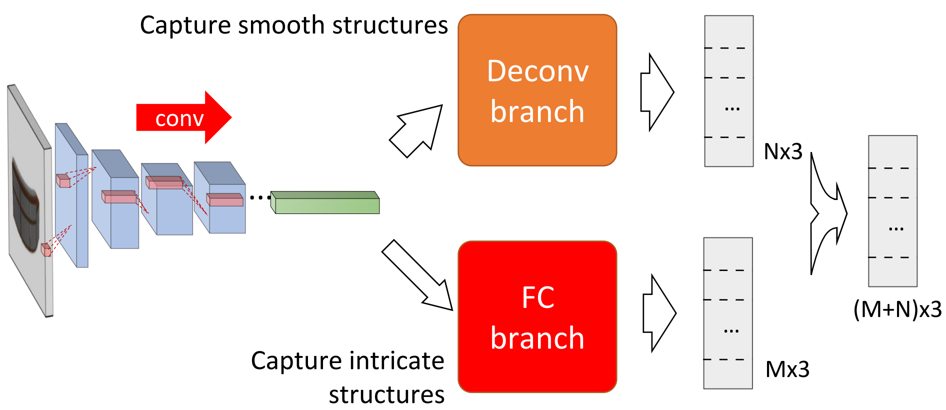
- Many local smooth structures are common
  - e.g., planar patches, cylindrical patches
  - **strong local correlation** among point coordinates

Natural Statistics of Object Geometry



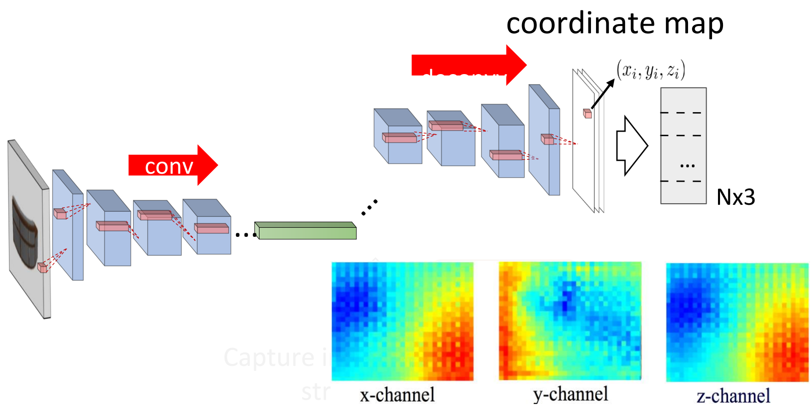
- But also some sharp/intricate local structures
  - **some points have high variability neighborhoods**

Two-Branch Architecture



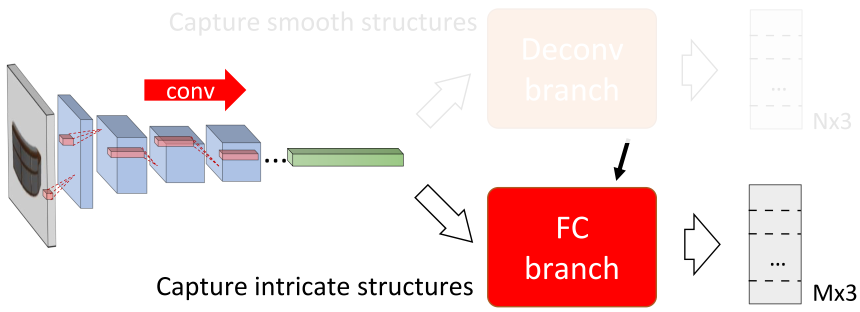
Set union by array concatenation

Deconvolution Branch



- Deconvolution induces a smooth coordinate map
- Geometrically, it learns a smooth parameterization

Fully-Connected Branch

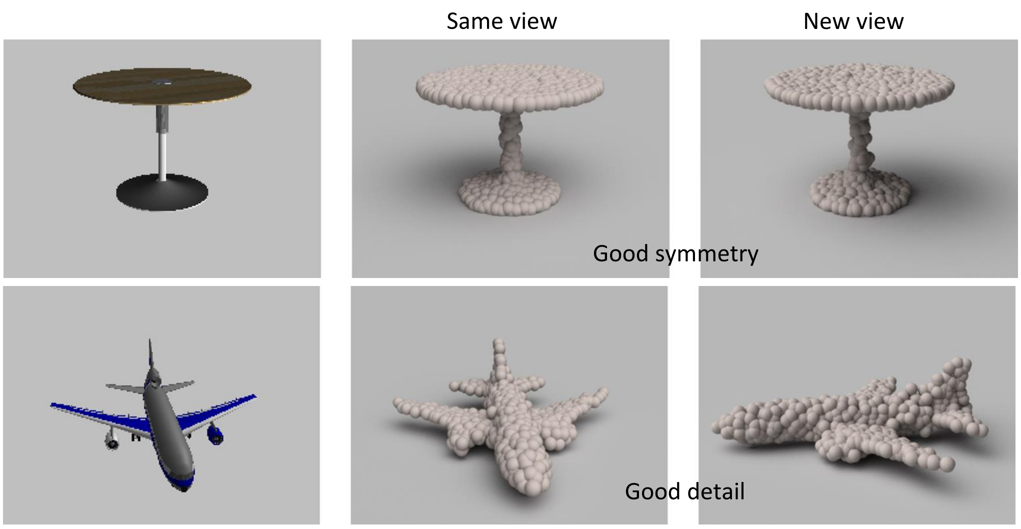


The Two Branches

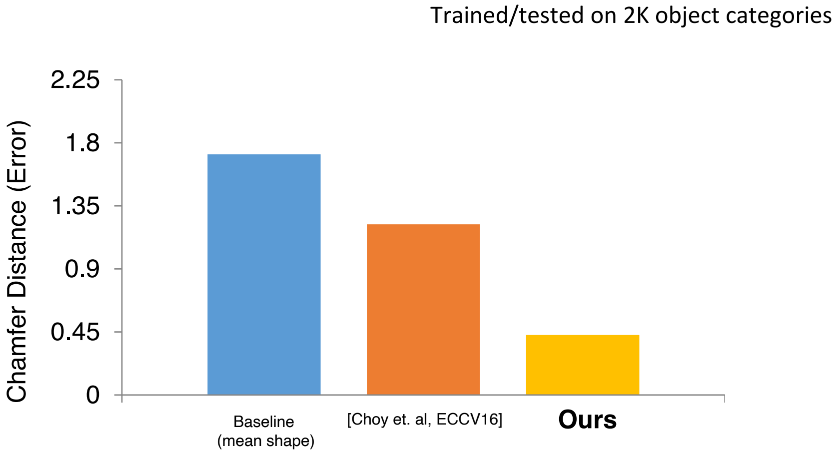
blue: deconv branch – **large, consistent, smooth** structures

red: fully-connected branch – **more intricate** structures

Example Results

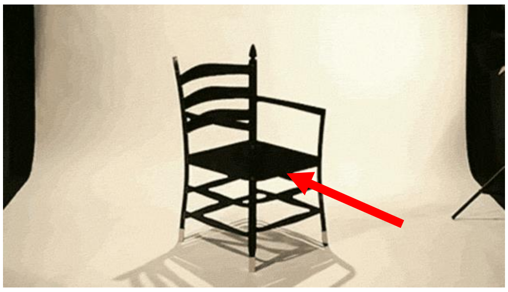


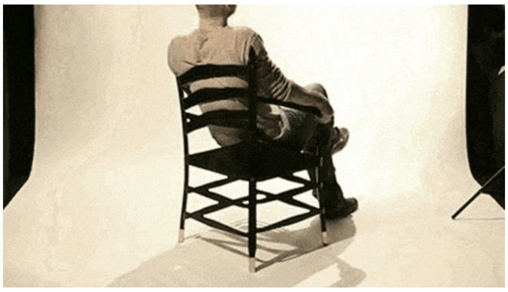
Comparison to State-of-the-Art



Influence of Distance Metrics

A fundamental issue: inherent ambiguity in prediction



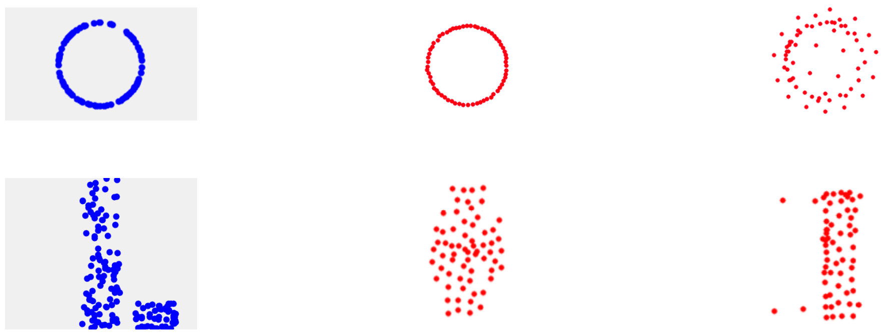


- By loss minimization, the network tends to predict a "**mean shape**" that **averages out** uncertainty

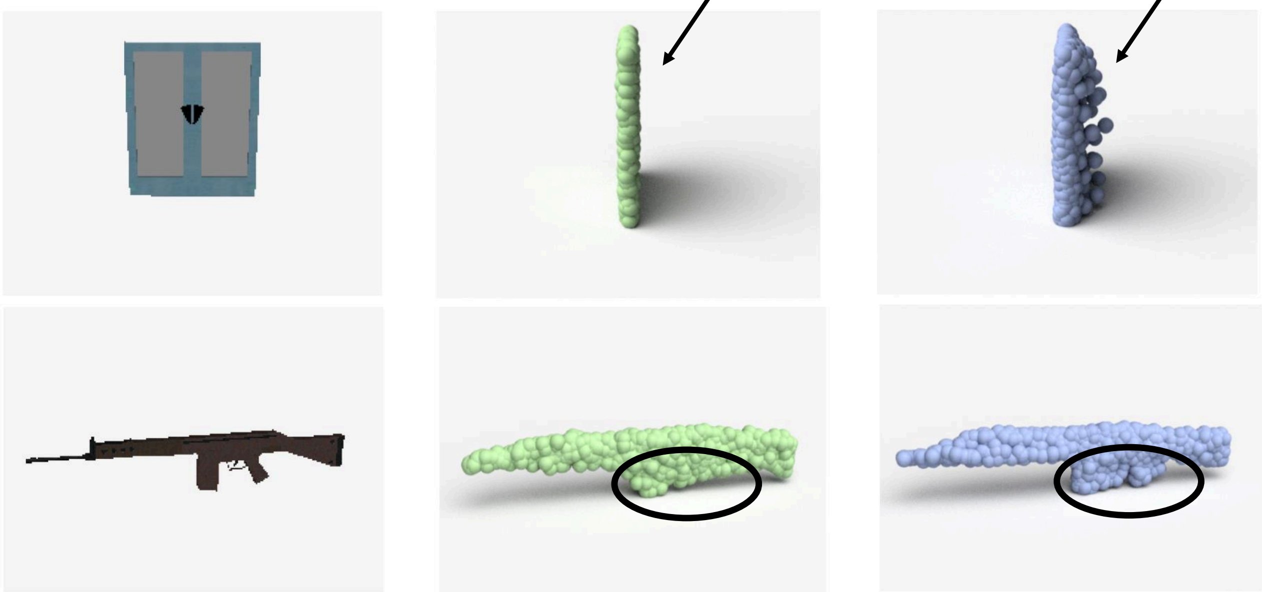
Distance Metrics Affect Mean Shapes

The mean shape carries characteristics of the distance metric

$$\bar{x} = \underset{x}{\operatorname{argmin}}\ \mathbb{E}_{s \sim \mathbb{S}}[d(x, s)]$$



Comparison of Predictions by EMD versus CD



From Real Images